

Question Three

(a) The pulley of an industrial electric motor has an angular acceleration of $\alpha = (1.5\theta) \text{ rad/s}^2$, where θ is in radians. The pulley has a radius of 0.2 m and assumed to start from rest. Determine:

- (i) the magnitude of the velocity and
- (ii) the acceleration of a point P located on the edge of the pulley after it has rotated 2 revolutions.

[15 Marks]

(b) If the pulley in Q. 3 (a) drives a wheel of 0.5 m diameter. Determine:

- (i) the speed of the wheel Q in revolution per minute (rpm) and
- (ii) the acceleration at a point m located at the outermost part of the wheel Q .

[10 Marks]

Question Four

(a) The motor winds in the cable with a constant acceleration, such that the 20-kg crate shown in Figure Q. 4 (a) moves a distance $s = 6 \text{ m}$ in 3 s , starting from rest. The coefficient of kinetic friction between the crate and the plane is $\mu_k = 0.3$. Determine:

- (i) the tension developed in the cable,
- (ii) the tension developed in the cable if the plane was completely horizontal, and
- (iii) the tension developed in the cable if the plane was completely vertical.

[15 Marks]

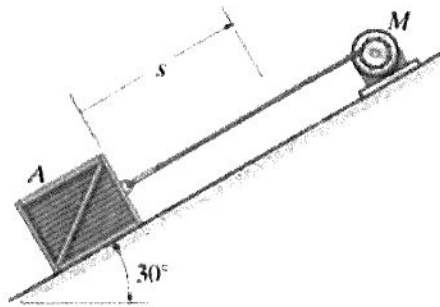


Figure Q. 4 (a)

(b) For the 20-kg crate shown in Figure Q. 4 (a), determine:

- (i) the acceleration and velocity of the crate at the inclined angle of 30° , and
- (ii) the acceleration and velocity of the crate if the plane was completely horizontal.

[10 Marks]

Question Five

(a) A car A has a mass of $2,000 \text{ kg}$ and a center of mass at G . The distance of the front and rear wheels from G are 1.25 m and 0.75 m respectively while the height of G is 0.3 m . If the mass of the wheels are neglected, and the coefficient of kinetic friction between the wheels and the road is $\mu_k = 0.25$. Determine:

- (i) the acceleration and
- (ii) the reactions at the front and rear wheels, if the rear "driving" wheels are always slipping, whereas the front wheels are free to rotate.

[13 Marks]

(b) Assuming a different car B was designed to function conversely to the car A in Question 5 (a). If the front "driving" wheels of car B are always slipping, whereas the rear wheels are free to rotate. Determine:

- (i) the acceleration, and
- (ii) the reactions at the front and rear wheels, assuming all other specifications of car B are similar to those of car A .

[12 Marks]